

Forecasting LA Home Values

Mortgage rates more than doubled after 2021. Did LA home values get harder to predict?

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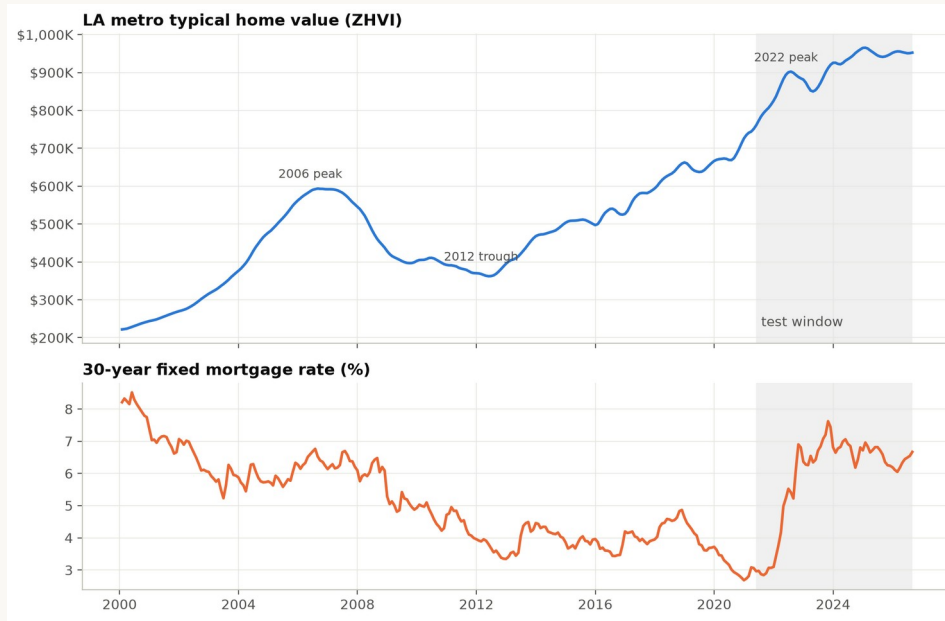
Zillow Home Value Index, LA metro, Jan 2000 to Aug 2026, plus 499 neighborhoods and the 30-year mortgage rate



Solid: actual · Dashed: our forecast to Aug 2027

THE QUESTION

LA values grew 5.6% a year for 26 years, then stalled when rates hit 7%



\$952,601

typical LA-metro home, August 2026

-1.3%

vs the January 2025 peak of \$965,634

2.7% → 7.6%

30-year mortgage rate, 2021 low to 2023 high

We tested 13 models two ways, so no model could win by luck

DATA

Zillow ZHVI, LA metro

320 months, Jan 2000 to Aug 2026

499 LA neighborhoods

Zillow neighborhood index

30-year mortgage rate

FRED, monthly

13 MODELS

Baselines

Naive drift

Classical

ARIMA · Holt · Prophet ×2

ML, deep learning, AI

XGBoost · LightGBM · LSTM · GRU ·
Chronos

Outside information

ARIMAX (rates) · VAR · factor VAR

2 TESTS

Blind 80/20

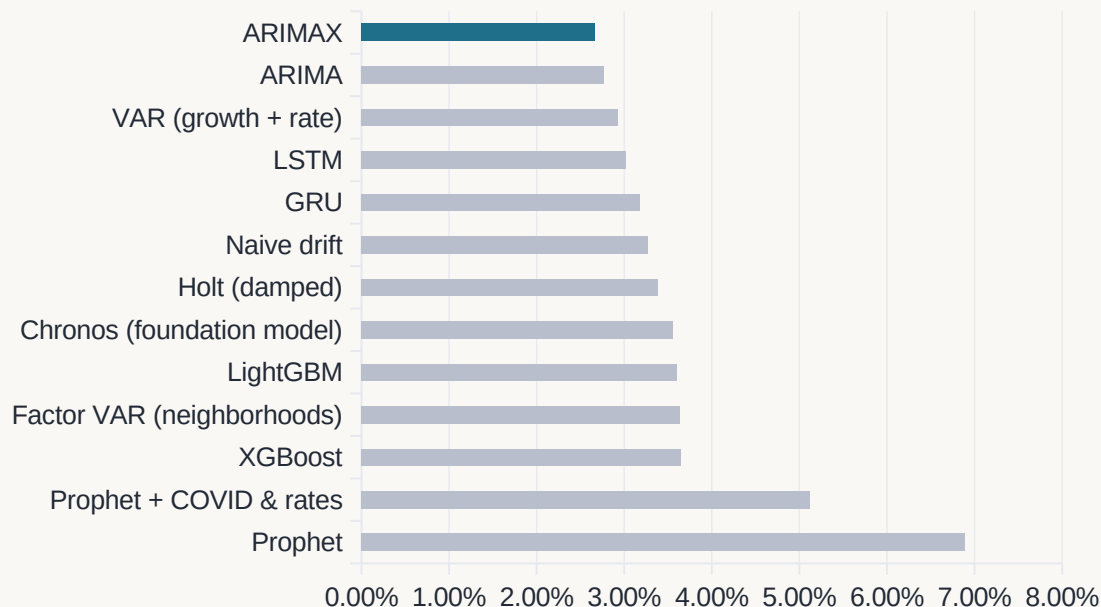
Train to Apr 2021, forecast 64 months in
one shot

Rolling origin

Refit every 3 months, forecast 12 months
ahead, 18 times

WHY NOT 1-MONTH ERRORS Zillow smooths this index, so every decent model is about 0.1% off one month ahead. We judge models at 6 to 12 months instead.

In the fair test, ARIMA-family models beat machine learning

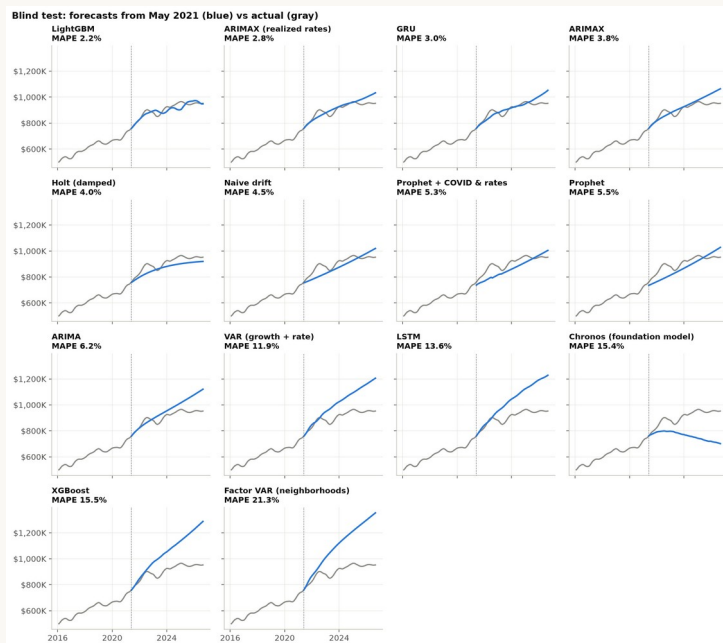


LOWEST ERROR

2.67%

ARIMAX, average error 1 to 12 months ahead. ARIMA 2.77%. LSTM 3.02%, XGBoost 3.65%, Prophet 6.89%.

One test window crowned LightGBM; 18 windows ranked it ninth

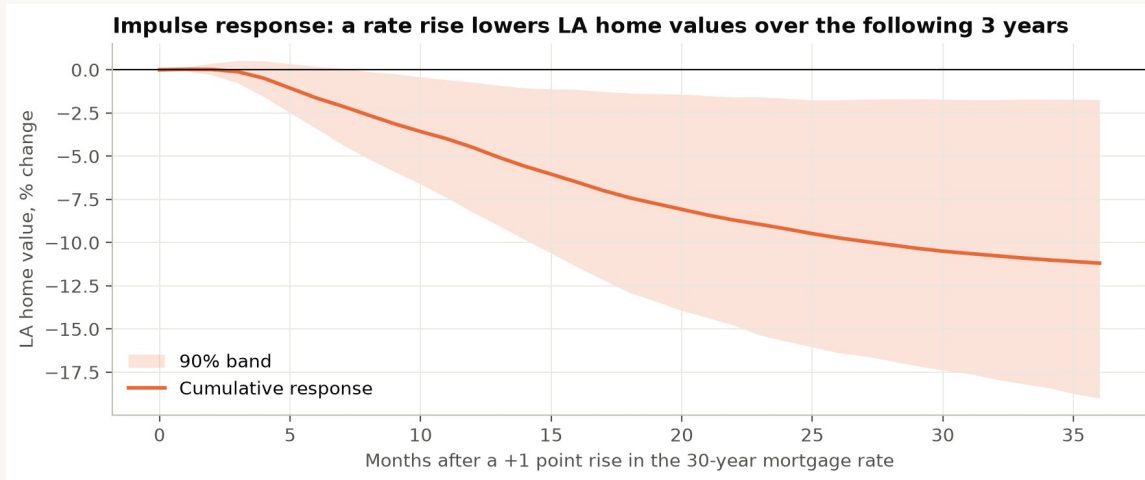


SAME MODEL, TWO TESTS

1st → 9th

LightGBM won the single 64-month test (2.2% error) but ranked 9th across 18 rolling tests: it got lucky in the one window that counted.

A 1-point rate rise means LA values about 9% lower two years later

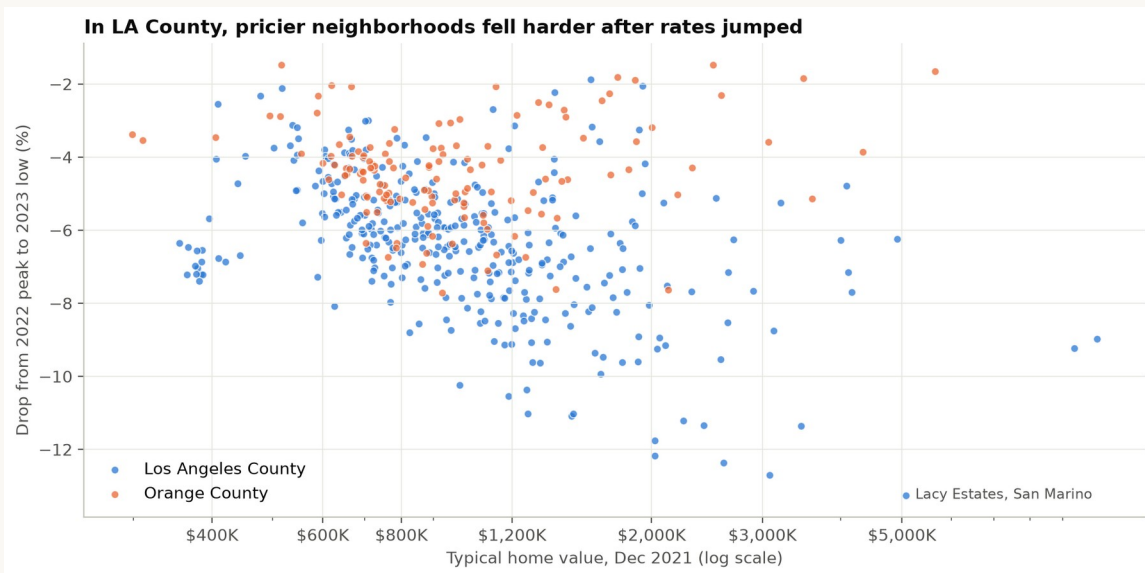


RATES WORK SLOWLY

-9.2%

after 24 months (-4.5% after 12). Rates help predict prices, not the reverse (Granger $p = 0.05$ vs 0.51).

In LA County, pricier neighborhoods fell hardest after rates jumped



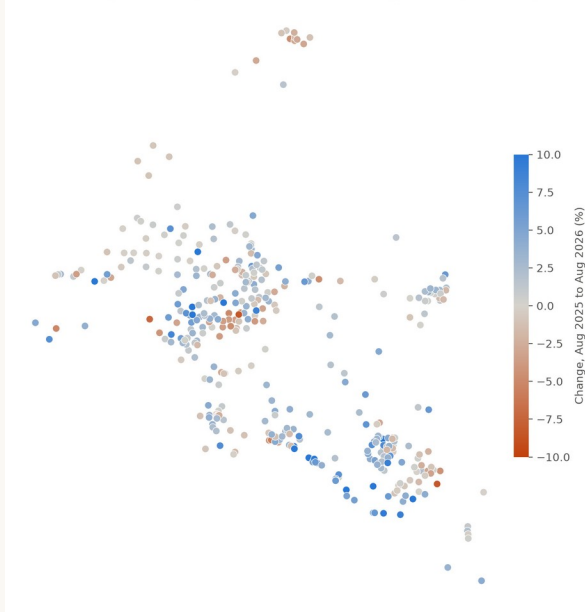
STILL BELOW 2022 PEAK

152 of 496

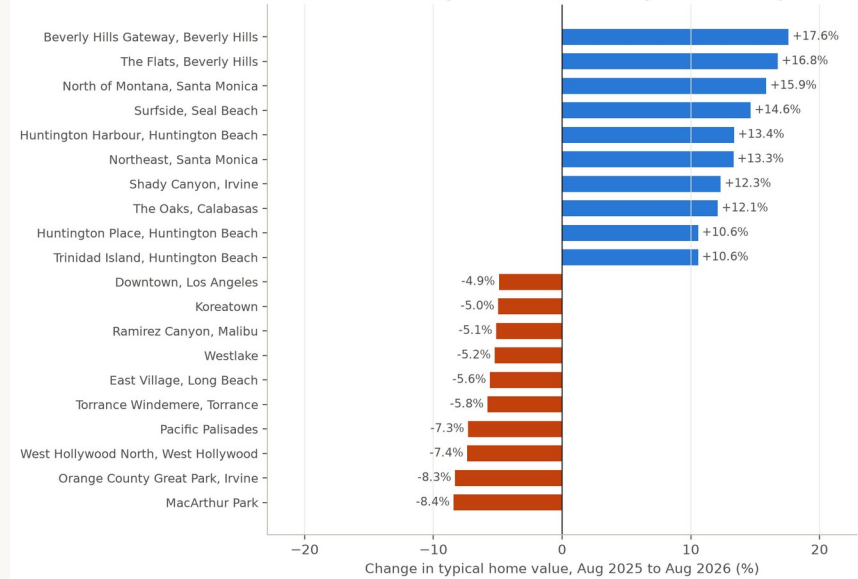
neighborhoods are still below their 2022 peak. In LA County, doubling the price meant a 1.2-point deeper drop ($t = -7.7$); in Orange County the effect reverses.

Right now coastal neighborhoods are rising while central LA falls

LA-metro neighborhoods, last 12 months: 220 rising, 216 flat, 60 falling

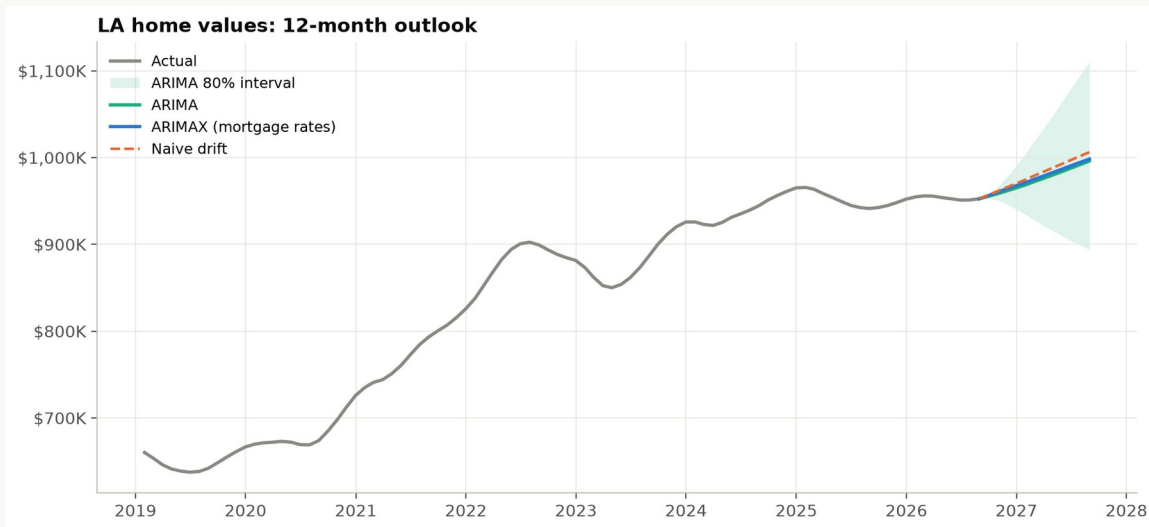


The 10 fastest-rising and 10 fastest-falling LA-metro neighborhoods



OUTLOOK

Our forecast: about 5% higher by August 2027, with a wide range



\$998,391

ARIMAX forecast, Aug 2027 (+4.8%)

-6% to +17%

80% range for the next 12 months

+4.8% / +6.5%

median neighborhood forecast, LA County
/ Orange County

What we learned

01

Simple beats complex for a smooth monthly index

ARIMA-family models beat LSTM, GRU, boosted trees and the Chronos foundation model.

02

How you test decides who wins

One blind window ranked LightGBM 1st; 18 rolling windows, 9th.

03

Mortgage rates matter, slowly

A 1-point rise lowers values about 9% over two years.

04

Use the long history

Training from 2000 beat starting in 2008, 2012 or 2016.

05

The metro average hides a split market

Separate models per neighborhood beat one shared factor (2.8% vs 4.1% error).